

# Physics-informed AI for Fast Three-phase N-1 Security Screening in Unbalanced Microgrids

Student Name, Advisor Name

**Abstract**—This mini-paper studies fast static N-1 security screening for unbalanced microgrids. A pandapower-based three-phase power-flow pipeline is used to generate post-contingency labels under voltage, loading, voltage-unbalance, service-loss, and convergence criteria. Several screening models are compared, including traditional ML baselines, a physics-informed multi-task MLP, and a phase-aware GNN. The goal is not to replace three-phase power flow, but to prioritize high-risk contingencies while maintaining high recall.

## I. Introduction

Unbalanced microgrids and low-voltage feeders can experience phase-specific voltage, current, and unbalance violations under component outages. Exhaustive three-phase N-1 scans are accurate but expensive when many operating scenarios and contingencies must be assessed. This work evaluates AI-based pre-screening models for fast high-recall contingency prioritization.

Contributions:

- A reproducible three-phase N-1 labeling workflow for microgrid security screening.
- A phase-aware violation definition including voltage, loading, VUF, service-loss, and non-convergence labels.
- A comparison of engineering rules, ML baselines, PI-MLP, and phase-aware GNN models under high-recall screening metrics.
- A validation suite for dataset lineage, no-leakage feature design, metric consistency, and graph invariance checks.

## II. Three-phase Microgrid Security Formulation

For each base-case scenario  $x_t$  and contingency  $c \in \mathcal{C}$ , the target is a binary label  $y_{t,c}$  and severity score  $s_{t,c}$ . The post-contingency quantities include phase voltages  $V_{i,\phi}^c$ , line currents  $I_{\ell,\phi}^c$ , and voltage unbalance factors.

### A. Violation Labels

The final label is the logical OR of low-voltage, high-voltage, loading, VUF, service-loss, critical service-loss, and non-convergence flags.

## III. Dataset Generation

The dataset is generated using a microgrid-like teaching feeder. Scenario generation controls load level, phase imbalance, PV penetration, and BESS charging/discharging setpoints. Each scenario is combined with a contingency catalog including line outages, PV trips, BESS trip, and PCC outage.

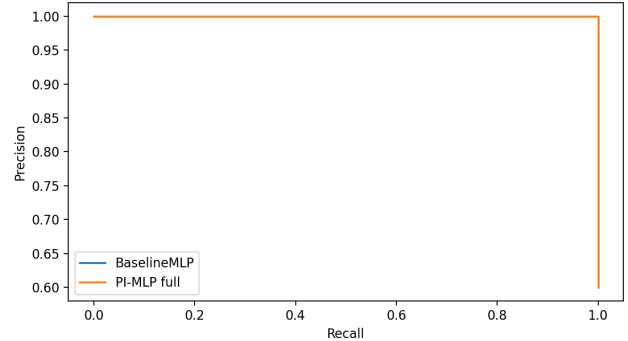


Fig. 1. Precision-recall curves for screening models.

## IV. Screening Models

### A. Traditional ML Baselines

Engineering rules, logistic regression, random forest, gradient boosting, and MLP baselines are trained using no-leakage base-case features.

### B. Physics-informed MLP

The PI-MLP predicts a violation probability, severity score, and component margins. The loss combines classification, severity, component-margin, component-severity consistency, probability-severity consistency, and ranking terms.

### C. Phase-aware GNN

The GNN uses bus-level phase-channel node features, line edge features, and contingency masks to produce a graph-level risk score.

## V. Experiments

### VI. Discussion and Limitations

The current benchmark is intentionally small and teaching-oriented. It supports methodological development and validation logic, but broader claims require additional feeders, stochastic scenarios, islanded-mode modeling, grid-forming inverter models, and real microgrid data.

## VII. Conclusion

This project demonstrates a reproducible workflow for three-phase microgrid N-1 security screening with AI. The main lesson is that high-recall screening, validation discipline, and no-leakage feature design are as important as model architecture.

TABLE I  
Dataset statistics for the teaching microgrid N-1 benchmark.

| dataset                           | n_scenarios | n_contingencies | expected_samples | actual_samples | safe_samples | unsafe_samples | power_flow_solved_rows | service_loss_skipped_rows | no_slack_skipped_rows |
|-----------------------------------|-------------|-----------------|------------------|----------------|--------------|----------------|------------------------|---------------------------|-----------------------|
| Three-phase microgrid N-1 dataset | 7           | 11              | 77               | 77             | 30           | 47             | 35                     | 35                        | 7                     |

TABLE II  
Holdout comparison of security screening models.

| stage              | model              | recall | precision | fnr   | pf_calls_saved_fraction | missed_violations | auprc | auroc |
|--------------------|--------------------|--------|-----------|-------|-------------------------|-------------------|-------|-------|
| Week 5 ML baseline | GradientBoosting   | 1.000  | 1.000     | 0.000 | 0.400                   | 0                 | 1.000 | 1.000 |
| Week 6 PI-MLP      | PI-MLP full        | 1.000  | 1.000     | 0.000 | 0.400                   | 0                 | 1.000 | 1.000 |
| Week 7 GNN         | FlattenedMLP       | 1.000  | 1.000     | 0.000 | 0.400                   | 0                 | 1.000 | 1.000 |
| Week 5 ML baseline | EngineeringRule    | 1.000  | 0.857     | 0.000 | 0.300                   | 0                 | 0.910 | 0.906 |
| Week 6 PI-MLP      | BaselineMLP        | 1.000  | 0.857     | 0.000 | 0.300                   | 0                 | 1.000 | 1.000 |
| Week 5 ML baseline | AllUnsafe          | 1.000  | 0.600     | 0.000 | 0.000                   | 0                 | 0.600 | 0.500 |
| Week 5 ML baseline | MLP                | 1.000  | 0.600     | 0.000 | 0.000                   | 0                 | 0.533 | 0.240 |
| Week 5 ML baseline | LogisticRegression | 0.833  | 1.000     | 0.167 | 0.500                   | 2                 | 1.000 | 1.000 |
| Week 5 ML baseline | RandomForest       | 0.833  | 0.909     | 0.167 | 0.450                   | 2                 | 0.956 | 0.938 |
| Week 7 GNN         | PhaseAwareGNN      | 0.583  | 0.875     | 0.417 | 0.600                   | 5                 | 0.913 | 0.875 |

TABLE III  
Group-based cross-validation for unseen scenarios and contingencies.

| stage              | model              | cv_type           | folds | mean_recall | mean_fnr | mean_pf_calls_saved | mean_auprc | missed_violations_sum |
|--------------------|--------------------|-------------------|-------|-------------|----------|---------------------|------------|-----------------------|
| Week 5 ML baseline | EngineeringRule    | scenario_group    | 3     | 1.000       | 0.000    | 0.323               | 0.912      | 0.000                 |
| Week 5 ML baseline | AllUnsafe          | scenario_group    | 3     | 1.000       | 0.000    | 0.000               | 0.596      | 0.000                 |
| Week 5 ML baseline | LogisticRegression | scenario_group    | 3     | 0.928       | 0.072    | 0.439               | 0.990      | 5.000                 |
| Week 5 ML baseline | RandomForest       | scenario_group    | 3     | 0.899       | 0.101    | 0.475               | 0.990      | 7.000                 |
| Week 5 ML baseline | EngineeringRule    | contingency_group | 3     | 1.000       | 0.000    | 0.310               | 0.971      | 0.000                 |
| Week 5 ML baseline | AllUnsafe          | contingency_group | 3     | 1.000       | 0.000    | 0.000               | 0.619      | 0.000                 |
| Week 5 ML baseline | RandomForest       | contingency_group | 3     | 0.400       | 0.600    | 0.690               | 0.861      | 28.000                |
| Week 5 ML baseline | LogisticRegression | contingency_group | 3     | 0.254       | 0.746    | 0.786               | 0.909      | 35.000                |
| Week 6 PI-MLP      | PI-MLP full        | contingency_group | 3     | 0.760       | 0.240    | 0.413               | 0.877      | NaN                   |
| Week 6 PI-MLP      | PI-MLP full        | scenario_group    | 3     | 0.928       | 0.072    | 0.207               | 0.942      | NaN                   |
| Week 7 GNN         | PhaseAwareGNN      | contingency_group | 11    | 0.896       | 0.104    | 0.247               | NaN        | 8.000                 |
| Week 7 GNN         | PhaseAwareGNN      | scenario_group    | 7     | 0.690       | 0.310    | 0.468               | NaN        | 18.000                |

TABLE IV  
Loss ablation for physics-informed MLP training.

| threshold | TP | FP | TN | FN | recall | precision | fnr   | pf_calls_saved_fraction | missed_violations | auprc | auroc | model          | selected_threshold |
|-----------|----|----|----|----|--------|-----------|-------|-------------------------|-------------------|-------|-------|----------------|--------------------|
| 0.010     | 12 | 2  | 6  | 0  | 1.000  | 0.857     | 0.000 | 0.300                   | 0                 | 1.000 | 1.000 | BaselineMLP    | 0.010              |
| 0.005     | 12 | 1  | 7  | 0  | 1.000  | 0.923     | 0.000 | 0.350                   | 0                 | 1.000 | 1.000 | PI-MLP no-rank | 0.005              |
| 0.005     | 12 | 0  | 8  | 0  | 1.000  | 1.000     | 0.000 | 0.400                   | 0                 | 1.000 | 1.000 | PI-MLP full    | 0.005              |

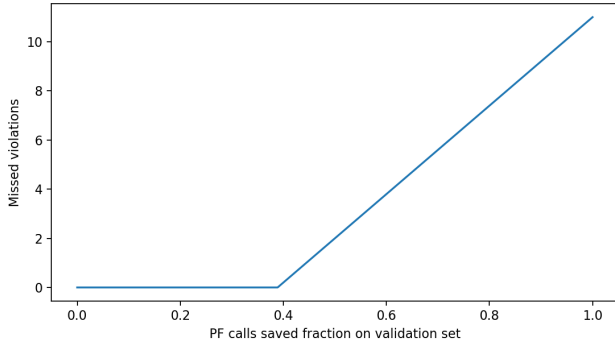


Fig. 2. Tradeoff between saved three-phase power-flow calls and missed violations.

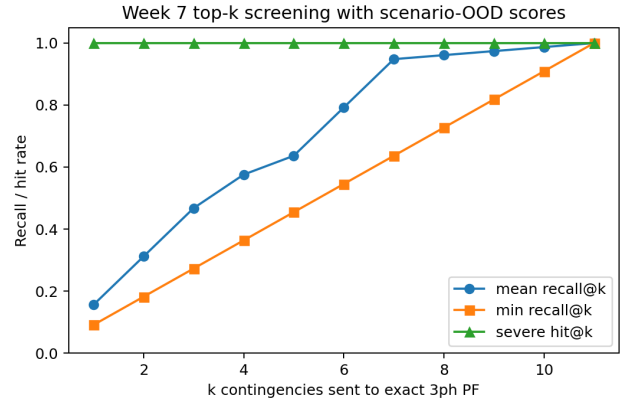


Fig. 3. Top-k violation recall for contingency prioritization.